

2.1 Measure Preliminaries-1

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* Let X be a non empty set. An algebra of sets of X is a non-empty collection $\mathcal{A}$ of subsets of X closed under finite union and complement.

A σ -algebra on X is a collection $\mathcal{M}$ of subsets of X satisfying

- (ci) $\emptyset \in \mathcal{M}$
- (ii) If $E \in \mathcal{M}$ then $E^c = X \setminus E \in \mathcal{M}$.
- (ciii) If $\{E_n\}_{n \in \mathbb{N}}$ is a seqⁿ of sets in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$.

Now $(X, \mathcal{M})$ is a measurable space and elements of $\mathcal{M}$ are called measurable set.

Properties of σ -algebra.

- 1) $X \in \mathcal{M}$
- 2) If $E_1, \dots, E_N \in \mathcal{M}$ then $\bigcup_{i=1}^N E_i \in \mathcal{M}$
- 3) If $E_1, \dots, E_N \in \mathcal{M}$ then $\bigcap_{i=1}^N E_i \in \mathcal{M}$
- 4) If $\{E_i\}_{i \in \mathbb{N}}$ is a seqⁿ of sets in $\mathcal{M}$, then $\bigcap_{i=1}^{\infty} E_i \in \mathcal{M}$
- 5) If $E, F \in \mathcal{M}$ then $E \setminus F \in \mathcal{M}$.

Proof:

- 1) $\emptyset \in \mathcal{M}$ and $X = (\emptyset)^c \in \mathcal{M}$.
- 2) Let $\{E_i\}_{i \in \mathbb{N}}$ be a countable collection of measurable sets where $E_i = \emptyset$ for $i > N$. Hence, $\bigcup_{i=1}^{\infty} E_i = E_1 \cup \dots \cup E_N \in \mathcal{M}$.
- 4) $(\bigcup_{i=1}^{\infty} E_i)^c = \bigcap_{i=1}^{\infty} E_i^c \in \mathcal{M}$
- 3) Apply 4) to collection of sets in 2)
- 5) $E \setminus F = E \cap F^c \in \mathcal{M}$ as finite intersection

is closed under σ -algebra.

****Proposition****

If $\{M_\alpha \mid \alpha \in J\}$ is collection of σ -algebra on X . Then $\bigcap_{\alpha \in J} M_\alpha$ is a σ -algebra.

****Proof:****

Observe $\emptyset \in M_\alpha$ for all $\alpha \in J$ this implies $\emptyset \in \bigcap_{\alpha \in J} M_\alpha$. If $E \in \bigcap_{\alpha \in J} M_\alpha$ then $E \in M_\alpha$ for all $\alpha \in J$ hence $E^c \in M_\alpha$ for all $\alpha \in J$ this implies $E^c \in \bigcap_{\alpha \in J} M_\alpha$. If $\{E_n\}_{n \in \mathbb{N}}$ is a seqⁿ of sets in $\bigcap_{\alpha \in J} M_\alpha$ hence $\bigcup_{n \in \mathbb{N}} E_n \in M_\alpha$ for each $\alpha \in J$. Hence $\bigcup_{n \in \mathbb{N}} E_n \in \bigcap_{\alpha \in J} M_\alpha$.

□

* In particular let $S \subseteq \mathcal{P}(X)$, there exists a unique smallest σ -algebra on X that contains S . This is called σ -algebra generated by S .

Remark 0.1. We show uniqueness by constructing a collection of σ -algebra containing S and then

taking an intersection over the family of σ -algebras. By the above proposition, the resulting collection is the smallest σ -algebra containing S .

* Let X be a topological space. The σ -algebra generated by topology on X is called Borel σ -algebra on X and denoted by B_X . The members of the σ -algebra are called Borel sets.

Note: F_σ is countable union of closed sets and G_δ is countable intersection of open sets. These are examples of Borel sets.

Proposition

$\mathcal{B}_{\mathbb{R}}$ is generated by each where $\mathcal{B}_{\mathbb{R}}$ is borel σ -algebra.

- a. open intervals $\mathcal{E}_1$
- b. closed intervals $\mathcal{E}_2$
- c. half-open intervals $\mathcal{E}_3$
- d. open rays $\mathcal{E}_4$
- e. closed rays $\mathcal{E}_5$

Proof: Observe $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_4$ and $\mathcal{E}_5$ are open or closed sets. For elements $\mathcal{E}_3$, these are G_δ sets i.e. $(a, b] = \bigcap_{n=1}^{\infty} (a, b + \frac{1}{n})$. Hence all are borel sets. If $\mathcal{M}(\mathcal{E}_j)$ is σ -algebra generated by $\mathcal{E}_j$. Then $\mathcal{M}(\mathcal{E}_j) \subseteq \mathcal{B}_{\mathbb{R}}$ for $j = 1, 2, 3, 4, 5$.

As any open set in $\mathbb{R}$ can be expressed as union of countable open intervals. Hence $B_{\mathbb{R}} \subseteq M(\mathcal{E})$.

Now observe for any open interval (a, b)

$$\begin{aligned} (a, b) &= \bigcap_{n=1}^{\infty} \left[a - \frac{1}{n}, b + \frac{1}{n} \right] = \left(a + \frac{1}{n} \right) \cap \left[a - \frac{1}{n}, b \right) \\ &= (a, \infty) \cap \bigcap_{n=1}^{\infty} \left(-\infty, b + \frac{1}{n} \right) \\ &= \bigcap_{n=1}^{\infty} \left[a - \frac{1}{n}, \infty \right) \cap (-\infty, b) \end{aligned}$$

Hence $M(\mathcal{E}_j) = B_{\mathbb{R}}$ for $j = 1, 2, 3, 4, 5$.

****Bonus!**** Every open set in $\mathbb{R}$ is a countable union of open intervals.

Let U be open set in $\mathbb{R}$. If $x \in U$ then x is rational or irrational. Let x be rational then construct $I_x = \bigcup_{x \in I} I$ where I are non disjoint open intervals containing x and contained in U . If x is not a rational for some $\epsilon > 0$, $x \in (a - \epsilon, a + \epsilon) \subseteq U$ choose a rational $q \in (a - \epsilon, a + \epsilon)$ and observe

$x \in I_q$. Hence for $x \in U$ there exists $q \in \mathbb{Q}$ st. $x \in I_q$. This implies $U \subseteq \bigcup_{q \in \mathbb{Q}} I_q$. Observe $I_q \subseteq U$ for all $q \in \mathbb{Q}$ hence

$$U = \bigcup_{q \in \mathbb{Q}} I_q \text{ where } I_q \text{ are open intervals.}$$

* A family of set $\mathcal{R} \subseteq \mathcal{P}(X)$ is called a ring if it closed under finite union and difference (i.e. if $E_1, \dots, E_n \in \mathcal{R}$ then $\bigcup_{i=1}^n E_i \in \mathcal{R}$ and if $E, F \in \mathcal{R}$ then $E \setminus F \in \mathcal{R}$). A ring that is closed under countable unions is called a σ -ring.

****Proposition****

If $\mathcal{R}$ is a σ -ring are closed under countable intersection.

****Proof:**** For $F \in \mathcal{R}$ observe $\emptyset = F \setminus F \in \mathcal{R}$. Let $\{E_i\}_{i=1}^\infty$ be a collection of sets belonging to $\mathcal{R}$. If any of them are pairwise disjoint $\bigwedge_{i=1}^n E_i = \emptyset \in \mathcal{R}$. Assume $\{E_i\}_{i=1}^\infty$ are not pairwise disjoint. As $\mathcal{R}$ is a σ -ring, let $\bigcup_{i=1}^n E_i = E \in \mathcal{R}$. Let $E'_i = E \setminus E_i \in \mathcal{R}$ for $i \in \mathbb{N}$. Now $E' = \bigcup_{i=1}^\infty E'_i \in \mathcal{R}$. Observe $\bigwedge_{i=1}^n E_i = E \setminus E' \in \mathcal{R}$.

* Let X be a set equipped with σ -algebra $\mathfrak{M}$

We define countably additive measure m such that

- (i) $m : \mathfrak{M} \rightarrow [0, \infty]$ is well defined function
- (ii) $m(\emptyset) = 0$.
- (iii) If $\{E_j\}_{j=1}^\infty$ is a seqⁿ of disjoint sets then

$$m\left(\bigcup_{j=1}^\infty E_j\right) = \sum_{j=1}^\infty m(E_j)$$

Observe (iii) implies finite additivity by taking $E_j = \emptyset$ for $j > n$.

* We define $(X, \mathfrak{M}, m)$ as a measure space.

If $m(X) < +\infty$ we call m as finite measure

If $X = \bigcup_{j=1}^\infty E_j$ where $E_j \in \mathfrak{M}$ and $m(E_j) < +\infty$ for all j then m is σ -finite measure.

More generally E is σ -finite set if $E = \bigcup_{j=1}^\infty E_j$

$E_j \in \mathfrak{M}$ and $m(E_j) < +\infty$ for all j .

* If for each $E \in \mathcal{M}$ with $m(E) = \infty$ and there exists $F \subseteq E$ st. $0 < m(F) < \infty$ then m is semifinite.

****Exercise****

Every σ -finite measure is semifinite.

****Proof:****

If $m(E) = \infty$ then observe $E = E \cap X = \bigcup_{j=1}^\infty (E \cap E_j)$ this implies $E \cap E_j \subseteq E_j \subseteq X$ for some j . We know $m(E_j) < +\infty$ hence $m(E \cap E_j) < m(E_j) < +\infty$. Hence we have found a subset of E specifically $E \cap E_j$ which has measure of $E_j \cap E$ to be finite.

Example 0.1. (i) Let X be any set and $x_0 \in X$ be a fixed point and $M = \mathcal{P}(X)$. Let $m : M \rightarrow \mathbb{R}$

$$m(E) = \begin{cases} 1 & \text{if } x_0 \in E \\ 0 & \text{if } x_0 \notin E \end{cases}$$

This is called the Dirac measure.

(ii) Let X be any set and $M = \mathcal{P}(X)$. Let $m : M \rightarrow [0, \infty]$

$$m(E) = \begin{cases} |E| & \text{if } |E| < +\infty \\ \infty & \text{otherwise} \end{cases}$$

where $|\cdot|$ is cardinality function and m is called the counting measure.

(iii) Let X be any set and $M = \mathcal{P}(X)$. Let $m : M \rightarrow [0, \infty]$

$$m(E) = +\infty \text{ for all } E \subset \mathbb{R}$$

Lemma 0.1. Suppose X be a set and $\mathfrak{M}$ is a σ -algebra equipped on X . Let $m : \mathfrak{M} \rightarrow [0, \infty]$ be a measure, then

(i) [Monotonicity] : If $A, B \in \mathfrak{M}$ s.t. $A \subseteq B$ then

$$m(A) \leq m(B).$$

(ii) If $\{A_n\} \subseteq \mathfrak{M}$ then

$$m\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} m(A_n).$$

Proof. (i) If $B \supseteq A$ then $m(B) = m(A \cup B \setminus A)$

$$= m(A) + m(B \setminus A) \geq m(A).$$

(ii) Let $E_1 = A_1$, and $E_k = A_k \setminus \bigcup_{i=1}^{k-1} A_i$ for $k \geq 1$. Then observe all E_k are disjoint to each other hence

$$m\left(\bigcup_{k \in \mathbb{N}} E_k\right) = \sum_{k \in \mathbb{N}} m(E_k)$$

then note

$$\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{k \in \mathbb{N}} E_k$$

and $m(E_k) \leq m(A_k)$ for all $k \in \mathbb{N}$. Hence we get

$$m\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{k \in \mathbb{N}} m(A_k).$$

□

Lemma 0.2. Let $(X, \mathcal{M}, \mu)$ be a measure space.

(a) Continuity from below: If $\exists E_j \uparrow_j^\infty = 1 \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \dots$, then $\mu(\bigcup_j E_j) = \lim_{j \rightarrow \infty} \mu(E_j)$.

(b) Continuity from above: If $\exists E_j \downarrow_j^\infty = 1 \subset \mathcal{M}$ and $E_1 \supset E_2 \supset E_3 \supset \dots$, then $\mu(\bigcap_j E_j) = \lim_{j \rightarrow \infty} \mu(E_j)$. and $\mu(E_1) < +\infty$.

Proof. (a) Set $E_0 = \emptyset$. Observe $\bigcup_j E_j = \bigcup_{j=1}^{\infty} E_j \setminus E_{j-1}$. Hence, $\mu(\bigcup_j E_j) = \mu(\bigcup_{j=1}^{\infty} E_j \setminus E_{j-1})$
 $= \sum_{j=1}^{\infty} \mu(E_j \setminus E_{j-1}) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu(E_j \setminus E_{j-1})$
 $= \lim_{n \rightarrow \infty} \mu(E_n)$.

(b) Let $F_j = E_1 \setminus E_j$. Observe $F_1 \subset F_2 \subset \dots$ and $\bigcup_{j=1}^{\infty} F_j = E_1 \setminus \bigcap_j E_j$.
 $\mu(E_1) = \mu(\bigcap_j E_j) + \lim_{n \rightarrow \infty} \mu(F_n)$ □

$$\begin{aligned} \mu(E_i) &= \mu\left(\bigcap_{j=1}^{\infty} E_j\right) + \lim_{n \rightarrow \infty} \mu(E_i \setminus E_j) \\ &= \mu\left(\bigcap_{j=1}^{\infty} E_j\right) + \mu(E_i) - \lim_{n \rightarrow \infty} \mu(E_j). \end{aligned}$$

This implies $\mu(\bigcap_{j=1}^{\infty} E_j) = \lim_{n \rightarrow \infty} \mu(E_j)$, as $\mu(E_i) < +\infty$, we can subtract it from both sides.

□

* If $(X, \mathcal{M}, \mu)$ is a measure space, a set $E \in \mathcal{M}$ such that $\mu(CE) = 0$ is called a null set.

If a statement is true about point $x \in X \setminus E$ for some null set E then we say the statement holds "almost everywhere" (a.e) (More precisely, we have μ -Null set and μ -a.e).

****Corollary****

Any countable union of null sets is a null set, and subsets of null sets belonging in $\mathcal{M}$ are null sets.

* If a measure whose domain include all subsets of a null set then the domain is called complete.

Theorem 0.1. Suppose that $(X, \mathcal{M}, \mu)$ is a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$ and $\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$. Then $\overline{\mathcal{M}}$ is a σ -algebra and there is a unique extension $\overline{\mu}$ of μ to complete measure $\overline{\mathcal{M}}$.

Proof. (a) $\overline{\mathcal{M}}$ is a σ -algebra

Since $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions, so is $\overline{\mathcal{M}}$. If $E \in \mathcal{M}$ and $F \subset N \in \mathcal{N}$ then $E \cup F \in \overline{\mathcal{M}}$. Now assume $E \cap N = \emptyset$ otherwise replace F by $F \setminus E$ and N by $N \setminus E$. Then $E \cup F = (E \cup N) \cap (N^c \cup F)$, so $(E \cup F)^c = (E \cup N)^c \cup (N^c \cup F)^c$ and observe

$(E \cup N)^c \in \mathcal{M}$ and $N \setminus F^c = N \setminus F \subset N$ and $N \setminus F \in \mathcal{N}$. Hence $(E \cup F)^c \in \overline{\mathcal{M}}$.

(cb) Define $\overline{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$, such that $\overline{\mu}(E \cup F) = \mu(CE)$. We want to show well definedness. Let $E_1 \cup F_1 = E_2 \cup F_2$ where $E_1, E_2 \in \mathcal{M}$ and $F_1, F_2 \in \mathcal{N}$. Now, $E_1 \subset E_1 \cup F_1 = E_2 \cup F_2 \subset E_2 \cup N_2$. This implies $\mu(CE_1) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$. If $E_2 \subset E_1 \cup N_1$, and this implies $\mu(E_2) \leq \mu(E_1)$. Hence $\mu(CE_1) = \mu(CE_2)$.

(c) Uniqueness. Let m_1, m_2 be extension of μ as per the above extension. Now, $E \cup F \in \overline{\mathcal{M}}$. Observe $m_1(E \cup F) = \mu(CE) = m_2(E \cup F')$ where F' is any subset of a null set.

* Outer measure

* An outer measure on a non-empty set X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ that satisfy,

* $\mu^*(\emptyset) = 0$.

* $\mu^*(A) \leq \mu^*(B)$ if $A \subseteq B$.

* $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$

****Proposition****

Let $E \subseteq \mathcal{P}(X)$ and $p : E \rightarrow [0, \infty]$ be such that $\emptyset \in E$, $X \in E$ and $p(\emptyset) = 0$. For any $A \subset X$ we define

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} p(E_j) \mid E_j \in E \text{ and } A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}$$

Then μ^* is an outer measure.

Proof: (a) Observe a countable collection of empty sets covers $A = \emptyset$ hence

$$\mu^*\emptyset = \inf \sum_{n \in \mathbb{N}} \rho(E_n) \leq \sum_{n \in \mathbb{N}} \rho(\emptyset) = 0$$

Also $\rho(E_n) \geq 0$ for any $E_n \in \mathcal{E}$ hence

$$\sum \rho(E_n) \geq 0$$

for any collection of intervals therefore

$$\inf \sum_{E_n \in \mathcal{E}} \rho(E_n) \geq 0.$$

So we get $\mu^*\emptyset = 0$.

For any non-empty set $A \subset X$ there exists $\{E_j\}_{j=1}^{\infty} \subset \mathcal{E}$ such that $A \subseteq \bigcup_{j=1}^{\infty} E_j$ we can set $E_j = X$ for all j .

(cb) If $A \subseteq B$ then if $\{E_j\}$ is collection of sets covering B then $\mathfrak{B}$ covers A as well.

If $\mathfrak{B}$ is a cover of B . We construct

$$A = \{E_n \in \mathfrak{B} \mid B_n \cap A \neq \emptyset\}$$

and observe $A \subseteq \mathfrak{B}$ Hence

$$\sum_{A_n \in A} P(A_n) \leq \sum_{B_n \in \mathfrak{B}} P(B_n)$$

therefore $\mu^*A \leq \sum_{B_n \in \mathfrak{B}} P(B_n)$. Observe $\mu^*A \leq \mu^*B$ otherwise $\mu^*A > \mu^*B$ then there exist a countable collection $\{E_j\}^2 = E$ covering B and also A whose sum i.e. $\sum P(B_n)$ is less than μ^*A which contradicts the fact that μ^*A is the infimum of sum of countable collection of sets in E covering A .

(c) Suppose $\{A_j\}_{j=1}^{\infty} \subset \mathcal{P}(X)$ and fix $\delta > 0$. For each $j \in \mathbb{N}$ we get $\{E_{j,k}\}_{k=1}^{\infty} \subset \mathcal{E}$ such that

$$A_j \subset \bigcup_{k=1}^{\infty} E_{j,k} \quad \text{and} \quad \sum_{k=1}^{\infty} \rho(E_{j,k}) \leq \mu^*(A_j) + \frac{\delta}{2^j}.$$

But if $A = \bigcup_{j=1}^{\infty} A_j$, we have $A \subset \bigcup_{j,k=1}^{\infty} E_{j,k}$ and

$$\sum_{j,k} \rho(E_{j,k}) \leq \sum_j \rho(E_{j,k}) + \delta, \quad \text{whence}$$

$$\mu^*(A) \leq \sum_j \mu^*(A_j) + \delta. \quad \text{As } \delta \text{ is arbitrary, we are done.}$$

□

* A set A is measurable if for each set E we have $m^*E = m^*(A \cap E) + m^*(A^c \cap E)$

Note: We know for any set $m^*E \leq m^*(A \cap E) + m^*(A^c \cap E)$ holds. Hence A is measurable iff

$$m^*E \geq m^*(A \cap E) + m^*(A^c \cap E) \quad \text{for any set } E.$$

Remark 0.2. If A is measurable then A^c is measurable.

Lemma 0.3. If A_1 and A_2 are measurable sets then $A_1 \cup A_2$ is measurable.

Proof. For any set $E \subset X$

$$\mu^*(E \cap (A_1 \cup A_2)) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2 \cap A_1^c)$$

and

$$\mu^*(E \cap (A_1 \cup A_2)) + \mu^*(E \cap (A_1 \cup A_2)^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2 \cap A_1^c) + \mu^*(E \cap A_2^c \cap A_1^c)$$

Now apply the fact that A_2 is a measurable set,

$$\mu^*(E \cap (A_1 \cup A_2)) + \mu^*(E \cap (A_1 \cup A_2)^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_2^c) = \mu^*(E)$$

as A_1 is a measurable set. Hence $A_1 \cup A_2$ is measurable. □

Lemma 0.4. Let E be any set, and $A_1, \dots, A_n$ a finite sequence of disjoint measurable sets. Then

$$\mu^* \left(E \cap \left(\bigcup_{i=1}^n A_i \right) \right) = \sum_{i=1}^n \mu^*(E \cap A_i)$$

Proof. Base case holds for $n = 1$. Assume the given holds for $n - 1$ sets.

Observe $E \cap \left(\bigcup_{i=1}^{n-1} A_i \right) \cap A_n = E \cap A_n$ and

$$E \cap \left(\bigcup_{i=1}^{n-1} A_i \right) \cap A_n^c = E \cap \bigcup_{i=1}^{n-1} A_i$$

as A_n is disjoint from $\xi_{i=1}^{n-1} A_i$. Hence

$$\begin{aligned} \mu^* \left(E \cap \left(\bigcup_{i=1}^n A_i \right) \right) &= \\ \mu^* \left(E \cap \bigcup_{i=1}^{n-1} A_i \cap A_n \right) &+ \mu^* \left(E \cap \bigcup_{i=1}^{n-1} A_i \cap A_n^c \right) \end{aligned}$$

by measurability of A_n . This implies

$$\mu^* \left(E \cap \left(\bigcup_{i=1}^n A_i \right) \right) = \mu^*(E \cap A_n) + \mu^* \left(E \cap \bigcup_{i=1}^{n-1} A_i \right)$$

□

By induction hypothesis's, we get

$$\mu^* \left(E \cap \left(\bigcup_{i=1}^n A_i \right) \right) = \sum_{i=1}^n \mu^*(E \cap A_i).$$

□

Theorem 0.2. (Carathéodory's Theorem) If μ^* is an outer measure on X , the collection $\mathcal{M}$ of μ^* -measurable sets is a σ -algebra and the restriction of μ^* to $\mathcal{M}$ is a complete measure.

Proof. According to a previous remark and lemma, $\mathcal{M}$ is an algebra and finitely additive. As per the previous lemma,

$$\mu^* \left(E \cap \left(\bigcup_{i=1}^{\infty} A_i \right) \right) = \sum_{i=1}^{\infty} \mu^*(E \cap A_i).$$

holds for any $E \subset X$ and collection of μ^* -measurable disjoint sets $\{A_i\}_{i=1}^{\infty}$. □

In order to $\mathcal{M}$ is closed under disjoint unions.

Let $B_n = \bigcup_{j=1}^n A_j$ and $B = \bigcup_{j=1}^{\infty} A_j$. Hence

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \\ &\geq \sum_{j=1}^n \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \end{aligned}$$

observe the statement above holds for any arbitrary n .

$$\mu^*(E) \geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$$

$$\begin{aligned}
&\geq \mu^* \left(\bigcup_{j=1}^{\infty} (E \cap A_j) \right) + \mu^*(E \cap B^c) \\
&= \mu^*(E \cap B) + \mu^*(E \cap B^c) \geq \mu^*(E).
\end{aligned}$$

It follows that $B \in \mathcal{M}$ and $\mu^*(B) = \sum_{j=1}^{\infty} \mu^*(A_j)$, so μ^* is countably additive, by substituting $E = B$.

Finally, if $\mu^*(A) < \infty$ for any $E \subset X$ we have

$$\begin{aligned}
\mu^*(E) &\leq \mu^*(E \cap A) + \mu^*(E \cap A^c) \\
&= \mu^*(E \cap A^c) \leq \mu(E).
\end{aligned}$$

so, $A \in \mathcal{M}$. Hence $\mu^*|_{\mathcal{M}}$ is complete measure

* Let $A \subset \mathcal{P}(X)$ is an algebra, a function $\mu_0 : A \rightarrow [0, \infty]$ will be called a premeasure if $\mu_0(\emptyset) = 0$,
* if $\{A_j\}_{j \in \mathbb{N}}$ is a sequence of disjoint sets in A , then $\mu_0(\bigcup_{j=1}^{\infty} A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$.

****Note:**** The difference between a measure and a premeasure is the domain.

* Measure is defined on a σ -algebra * Premeasure is defined on an algebra.

* We have a notion of finite and σ -finite premeasures as same as that of a measure.

Remark 0.3. If μ_0 is a premeasure on $\mathcal{A} \subseteq \mathcal{PCX}$, it induces an outer measure on X in accordance with Proposition /ref , namely

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(A_j) \mid A_j \in \mathcal{A}, E \subseteq \bigcup_{j=1}^{\infty} A_j \right\}$$

Lemma 0.5. If μ_0 is a premeasure on $\mathcal{A}$ and μ^* as defined in remark above, then

1. $\mu^*|_{\mathcal{A}} = \mu_0$;
2. every set in $\mathcal{A}$ is μ^* -measurable.

Proof. a) Suppose $E \in \mathcal{A}$. If $E \subseteq \bigcup_{j=1}^{\infty} A_j$ with any countable collection $\{A_j\}_{j=1}^{\infty} \subseteq \mathcal{A}$ then we let $B_n = E \cap (A_n \setminus \bigcup_{j=1}^{n-1} A_j)$. Observe B_n is a disjoint member of $\mathcal{A}$ whose union is E , so $\square$

$\mu_0(E) = \sum_{n=1}^{\infty} \mu_0(B_n) \leq \sum_{n=1}^{\infty} \mu_0(A_j)$. Since this holds for any arbitrary cover of E hence as per the definition of μ^* , we get $\mu_0(E) \leq \mu^*(E)$. Now as $E \in \mathcal{A}$, we can choose $A_j \subset \bigcup_{j=1}^{\infty} A_j$ st. $A_1 = E$ and $A_j = \emptyset$. Hence $\mu_0(E) = \mu^*(E)$.

b) If $A \in \mathcal{A}$, $E \subset X$ and $\epsilon > 0$ then there is a sequence $B_j \subset_{j \in \mathbb{N}} A$ with $E \subset \bigcup_{j=1}^{\infty} B_j$ and $\sum_{j=1}^{\infty} \mu_0(B_j) \leq \mu^*(E) + \epsilon$. Since μ_0 is additive on $\mathcal{A}$,

$$\mu_0(\bigcup_{j=1}^{\infty} B_j \cap X) = \mu_0(\bigcup_{j=1}^{\infty} B_j \cap (A \cup A^c))$$

$$\begin{aligned}
&= \mu_0(\bigcup_{j=1}^{\infty} (B_j \cap A)) + \mu_0(\bigcup_{j=1}^{\infty} (B_j \cap A^c)) \\
&= \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c).
\end{aligned}$$

Hence,

$$\begin{aligned}
\mu^*(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(B_j \cap A) + \sum_{j=1}^{\infty} \mu_0(B_j \cap A^c) \\
&\geq \mu_0(E \cap A) + \mu_0(E \cap A^c)
\end{aligned}$$

As ϵ is arbitrary, A is μ^* -measurable.

Theorem 0.3. Let $A \subseteq \mathcal{P}(X)$ be an algebra, μ_0 is a premeasure defined on A . Let $\mathcal{M}$ be σ -algebra generated by A . There exists a measure μ on $\mathcal{M}$ such that $\mu|_A = \mu_0$.

Proof. If we define μ^* as defined earlier and $\mu := \mu^*|_{\mathcal{M}}$. Then we know by the previous lemma $\mu^*|_A = \mu|_A = \mu_0$ and all A are μ^* -measurable. This implies σ -algebra of μ^* -measurable sets include A . $\square$

Theorem 0.4. If ν is another measure defined on $\mathcal{M}$ (as per the previous theorem) that extends on μ_0 then $\nu(E) \leq \mu(E)$ for any $E \in \mathcal{M}$ and equality when $\mu(E) < +\infty$. If μ_0 is σ -finite, then μ is the unique extension of μ_0 to a measure on $\mathcal{M}$.

Proof. If $E \in \mathcal{M}$ and $E \subset \bigcup_{j=1}^{\infty} A_j$ where $A_j \in \mathcal{A}$ for all $j \in \mathbb{N}$, then $\nu(CE) \leq \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j)$ holds true for any $\{A_j\}_{j=1}^{\infty}$ collection. Hence,

$$\nu(E) \leq \mu(CE).$$

Let $A = \bigcup_{j=1}^{\infty} A_j$. Now $\nu(A) = \nu\left(\bigcup_{j=1}^{\infty} A_j\right)$

$$\begin{aligned}
&= \sum_{j=1}^{\infty} \nu(A_j) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \nu(A_j) = \lim_{n \rightarrow \infty} \nu\left(\bigcup_{j=1}^n A_j\right) \\
&= \lim_{n \rightarrow \infty} \mu_0\left(\bigcup_{j=1}^n A_j\right) = \sum_{j=1}^{\infty} \mu_0(A_j) = \mu_0(A).
\end{aligned}$$

$\square$

If $\mu(E) < +\infty$. For every $\epsilon > 0$ there exist a collection of sets $\{A_{\epsilon, j}\}_{j=1}^{\infty}$, such that $A_{\epsilon} = \bigcup_{j=1}^{\infty} A_{\epsilon, j}$ such that $\mu(A_{\epsilon}) < \mu(CE) + \epsilon$, hence $\mu(A_{\epsilon} \setminus E) < \epsilon$, and $\mu(CE) \leq \mu(A) = \nu(A) = \nu(E) + \nu(A \setminus E) \leq \nu(CE) + \mu(A \setminus E) \leq \nu(CE) + \epsilon$ holds for all $\epsilon > 0$. This implies $\mu(CE) \leq \nu(CE)$.

Finally suppose $X = \bigcup_j A_j$ with $\mu_0(A_j) < \infty$ where we can assume A_j 's are disjoint (or reconstruct a given collection into that way). Then for any $E \in \mathcal{M}$

$$\begin{aligned}
\mu(E) &= \sum_j \mu(E \cap A_j) = \sum_j \nu(CE \cap A_j) \\
&= \nu(CE),
\end{aligned}$$

so $\mu = \nu$.

A: finite disjoint unions of h-intervals is an algebra.

Borel Measure on $\mathbb{R}$

Notation: $(a, b]$ are denoted as h-intervals.

μ is the borel measure on $\mathcal{B}_{\mathbb{R}}$. We define distribution function as $F(x) = \mu((-\infty, x])$

Proposition

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing and right cont. If $\{(a_j, b_j]\}_{j=1}^n$ are disjoint h-intervals, let

$$\mu_0 \left(\bigcup_{j=1}^n (a_j, b_j] \right) = \sum_{j=1}^n \mu_0((a_j, b_j]),$$

and let $\mu_0(\emptyset) = 0$. Then μ_0 is a premeasure on algebra A .

Proof: (a) μ_0 is well defined

Given any collection of disjoint h-intervals. If $\{(a_j, b_j]\}_{j=1}^n$ by relabelling, we get

$$a = a_1 < b_1 = a_2 < \dots < b_n = b.$$

This implies $\sum_{j=1}^n [F(b_j) - F(a_j)] = F(b) - F(a)$.

More generally, if $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^m$ are finite sequences of disjoint h -intervals such that $\bigcup_{i=1}^n I_i = \bigcup_{j=1}^m J_j$ by the reasoning above we can say,

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j).$$

(b) μ_0 is countably additive

Observe μ_0 is finitely additive by construction.

We want to show if $\sum I_i \subseteq \mathcal{U}$ st. $\bigcup_{i \in \mathbb{N}} I_i \in \mathcal{U}$ then

$$\mu_0 \left(\bigcup_{i=1}^{\infty} I_i \right) = \sum_{i=1}^{\infty} \mu_0(I_i).$$

As $\bigcup_{i=1}^{\infty} I_i \in \mathcal{U}$, we can separate $\{I_i\}_{i=1}^{\infty}$ into finitely many subsequences whose union is an h -interval.

Assume $\bigcup_{i=1}^{\infty} I_i$ is an h -interval (otherwise you look into a smaller collection of I_i 's whose union is an h -interval.)

$I = \bigcup_{i=1}^{\infty} I_i = [a, b)$. Observe

$$\mu_0(I) = \mu_0 \left(\bigcup_{j=1}^n I_j \right) + \mu_0 \left(I \setminus \bigcup_{j=1}^n I_j \right)$$

$$\geq \mu_0 \left(\bigcup_{j=1}^n I_j \right) = \sum_{j=1}^n \mu_0(I_j)$$

holds for any $n \in \mathbb{N}$, we obtain $\mu_0(I) \geq \sum_{j=1}^n \mu(I_j)$.

To show reverse inequality

case (ci): If a, b are finite.

As F is right continuous, for any $\epsilon > 0$, there exists $\delta_\epsilon > 0$ such that $F(a + \delta_\epsilon) - F(a) < \epsilon$.

If $I_j = (a_j, b_j]$ $j \in \mathbb{N}$ then we can find a $\delta_{\epsilon,j} > 0$ such that $F(b_j + \delta_{\epsilon,j}) - F(b_j) < \frac{\epsilon}{2^j}$.

Now $\{(a_j, b_j + \delta_{\epsilon,j})\}_{j=1}^\infty$ is a collection of open intervals covering $[a + \delta_\epsilon, b]$, so there is a finite subcover.

Let $(a_1, b_1 + \delta_1), \dots, (a_N, b_N + \delta_N)$ cover $[a + \delta_\epsilon, b]$ and $b_i + \delta_i \in (a_{i+1}, b_{i+1} + \delta_{i+1})$.

(We have replaced $\delta_{\epsilon,i} = \delta_i$ for all $i = \{1, \dots, N-1\}$).

For any $\epsilon > 0$, observe

$$\begin{aligned} \mu_0(I) &< F(b) - F(a + \delta_\epsilon) + \epsilon \\ &\leq F(b_N + \delta_N) - F(a_1) + \epsilon \\ &= F(b_N + \delta_N) - F(a_N) + \sum_{i=1}^{N-1} [F(a_{i+1}) - F(a_i)] + \epsilon \\ &\leq F(b_N + \delta_N) - F(a_N) + \sum_{i=1}^N [F(b_j + \delta_j) - F(a_j)] + \epsilon \\ &< \left(\sum_{i=1}^N F(b_j) + \frac{\epsilon}{2^j} - F(a_j) \right) + \epsilon < \sum_{i=1}^\infty \mu(I_j) + 2\epsilon. \end{aligned}$$

As the above statement holds for all $\epsilon > 0$, we are done.

****case (ii):**** If one of them is NOT finite.

Say $a = -\infty$. Fix $\epsilon > 0$.

Then for any $M > 0$, we can find an t open cover $(a_j + b_j + \delta_j)$ covering $[-M, b]$. Hence by the same reasoning we get

$$F(b) - F(-M) \leq \sum_{j=1}^\infty \mu_0(I_j) + 2\epsilon,$$

whereas if $b = \infty$, for any $M < \infty$ we get

$$F(M) - F(a) \leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon.$$

We get the desired result by $\varepsilon \rightarrow 0$ followed by $M \rightarrow \infty$. (Go through this).

Theorem 0.5. If $F : \mathbb{R} \rightarrow \mathbb{R}$ is any increasing, right continuous d^m , there is a unique Borel measure μ_F on $\mathbb{R}$ such that $\mu_F((a, b]) = F(b) - F(a)$ for all a, b . If G is another function similar to this then $G - F$ is constant iff $\mu_F = \mu_G$

Conversely if μ is a Borel measure on $\mathbb{R}$ that is finite on all bounded Borel sets and we define

$$F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((x, 0]) & \text{if } x < 0 \end{cases}$$

Proof. Each F induces a premeasure on $\mathcal{U}$ by previous proposition. It is clear that F and G induce the same premeasure iff $F - G$ is constant, and these premeasure are σ -finite as $\mathbb{R} = \bigcup_{j \in \mathbb{Z}} (j, j+1]$. $\square$

The first two assertions therefore follow from Theorem. As for the converse, the monotonicity of μ implies F to be an increasing function.

Observe μ is continuous from above and below implies the right continuity of F for $x \geq 0$ and $x < 0$. It is evident that $\mu = \mu_F$ on $\mathcal{U}$, and hence $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$ by uniqueness of Theorem.

Remark 0.4. $\circ$ The theory could have also been developed for $[a, b)$ intervals and F as left cont. S^n .

$\circ$ If μ is a finite Borel measure on $\mathbb{R}$ then $F(x) = \mu((-\infty, x])$ is the cumulative distribution function.

$\circ$ Any increasing right cont. F^n gives us a complete measure.

Given an increasing, right continuous function F . We get $(\mathbb{R}, \mathcal{M}_{\mu}, \mu)$ as a complete measure space where for any $E \in \mathcal{M}_{\mu}$,

$$\mu(E) = \inf \left\{ \sum_{j=1}^{\infty} [F(b_j) - F(a_j)] \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

Lemma 0.6. For any $E \in \mathcal{M}_{\mu}$,

$$\mu(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu((a_j, b_j)) \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

Proof. Let $\nu : \mathcal{M}_{\mu} \rightarrow [0, \infty)$ such that

$$\nu(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu((a_j, b_j)) \mid E \subset \bigcup_{j=1}^{\infty} (a_j, b_j] \right\}$$

and μ be the measure defined in above given defⁿ.

Goal : $\nu(E) = \mu(E)$ for all $E \in \mathcal{M}_\mu$. □

(i) We want to show $\mu(E) \leq \nu(E)$ for any $E \in \mathcal{M}_\mu$.

Suppose $E \subseteq \bigcup_{j=1}^{\infty} (a_j, b_j)$ and for each (a_j, b_j) we get $\{I_j^k\}_{k=1}^{\infty}$ as disjoint h -intervals covering (a_j, b_j) for each k . Let $I_j^k = [c_j^k, c_j^{k+1}]$ where $c_j^1 = a_j$ and $c_j^k \rightarrow b_j$ as $k \rightarrow \infty$. Thus $E \subseteq \bigcup_{j,k} I_{j,k}$. Hence

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) = \sum_{j=1}^{\infty} \mu(I_j^k) \geq \mu(E)$$

(Or you can use $E \subseteq \bigcup_{j \in \mathbb{N}} (a_j, b_j)$ and by monotonicity and subadditivity and apply infimum, we get $\mu(E) \leq \nu(E)$).

(ii) WTS : $\nu(E) \leq \mu(E)$; for any $\varepsilon > 0$, we get $\{(a_j, b_j)\}_{j=1}^{\infty}$ covering E and

$$\sum_{j=1}^{\infty} \mu((a_j, b_j)) \leq \mu(E) + \varepsilon.$$

As F is right cont. and increasing, for $\epsilon > 0$ we find $\delta_{\epsilon j} = \delta_j$ for each $j \in \mathbb{N}$ st.

$$F(b_j + \delta_j) - F(b_j) < \epsilon/2^j.$$

Then $E \subset \bigcup_{j=1}^{\infty} (a_j + b_j + \delta_j)$ and

$$\begin{aligned} \mu((a_j, b_j + \delta_j)) &\leq \mu((a_j, b_j + \delta_j)) \\ &= F(b_j + \delta_j) - F(a_j) \\ &= F(b_j + \delta_j) - F(b_j) + F(b_j) - F(a_j) \end{aligned}$$

for all $j \in \mathbb{N}$. Now by taking a sum

$$\begin{aligned} \sum_{j=1}^{\infty} \mu((a_j, b_j)) &\leq \sum_{j=1}^{\infty} F(b_j + \delta_j) - F(b_j) \\ &\quad + \sum_{j=1}^{\infty} \mu((a_j, b_j)) \\ &\leq \epsilon + \epsilon + \mu(E). \end{aligned}$$

Observe $\nu(E) \leq \sum_{j=1}^{\infty} \mu((a_j, b_j)) \leq \mu(E) + 2\epsilon$ holds for all $\epsilon > 0$. Hence $\nu(E) \leq \mu(E)$.

Theorem 0.6. If $E \in \mathcal{M}_\mu$, then

$$\begin{aligned} \mu(E) &= \inf \{ \mu(U) \mid U \supseteq E \text{ and } U \text{ is open} \} \\ &= \sup \{ \mu(K) \mid K \subseteq E \text{ and } K \text{ is compact} \} \end{aligned}$$